

Applying Pre-Stretching to Conductive Electrode Layer in DEAs for better Anisotropic Dielectric Elastomers (MATS)

All Images/Photographs Made By Researcher Unless Otherwise Mentioned.

Background & Introduction

- Research has seen growth specifically in soft robotics research.
- Soft robots are made of lightweight, inexpensive, flexible materials that allow them to adapt.

- One method for this is a three-layer **dielectric elastomer actuator (DEA)**.
 - The **dielectric layer** is made of a non-conductive polymer while the **electrode layers** are made from conductive polymers.
 - When a voltage is applied to the electrodes, the electrodes are attracted, applying stress onto the dielectric layer.

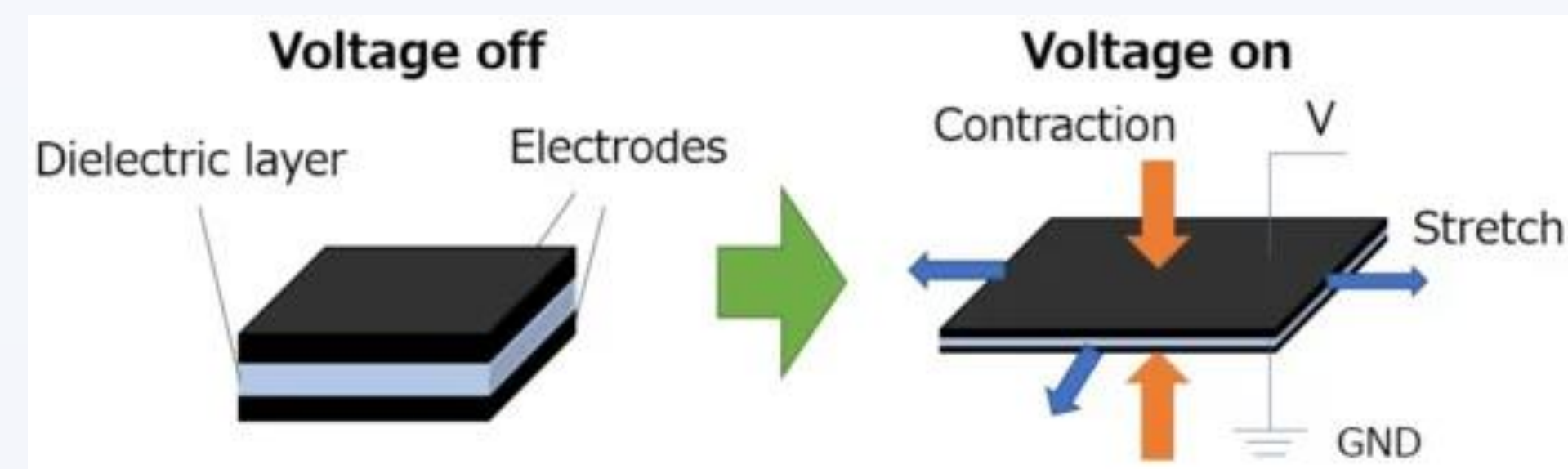


Figure 1. Diagram describing dielectric elastomer actuators.

- Anisotropic Dielectric Elastomers (ADE)** are a type of DEAs that focus on directional actuation.
 - ADEs have the same layers as a normal DEA but with **mechanical reinforcement** applied to the dielectric.
 - Methods for this include implementing nanofibers and pre-stretching the layers.

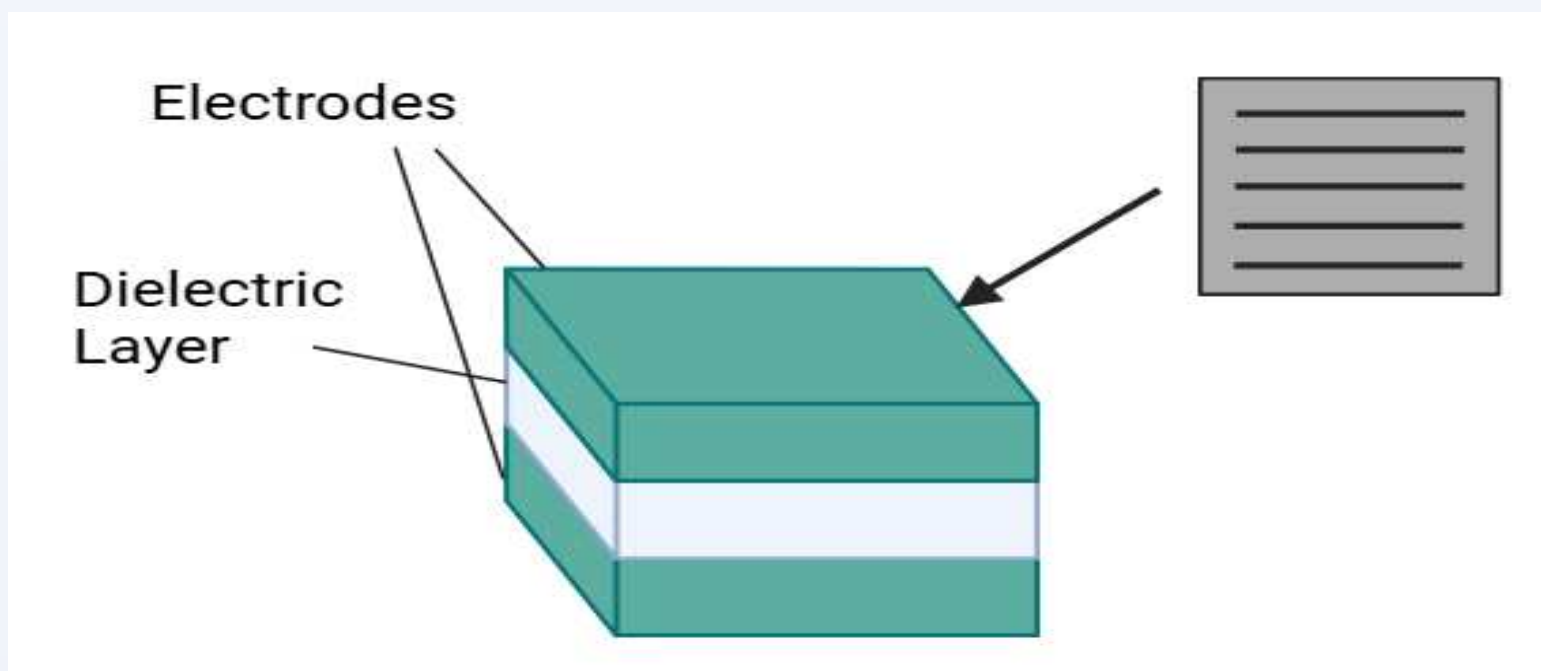


Figure 2. Diagram describing anisotropic elastomer actuators.

Objective & Variables

Research Question:

- How can I make a better functioning ADE by applying mechanical reinforcement in the electrode layer composite made using PDMS and Carbon Black (Graphite), and will it perform better than current ADEs.

Variables:

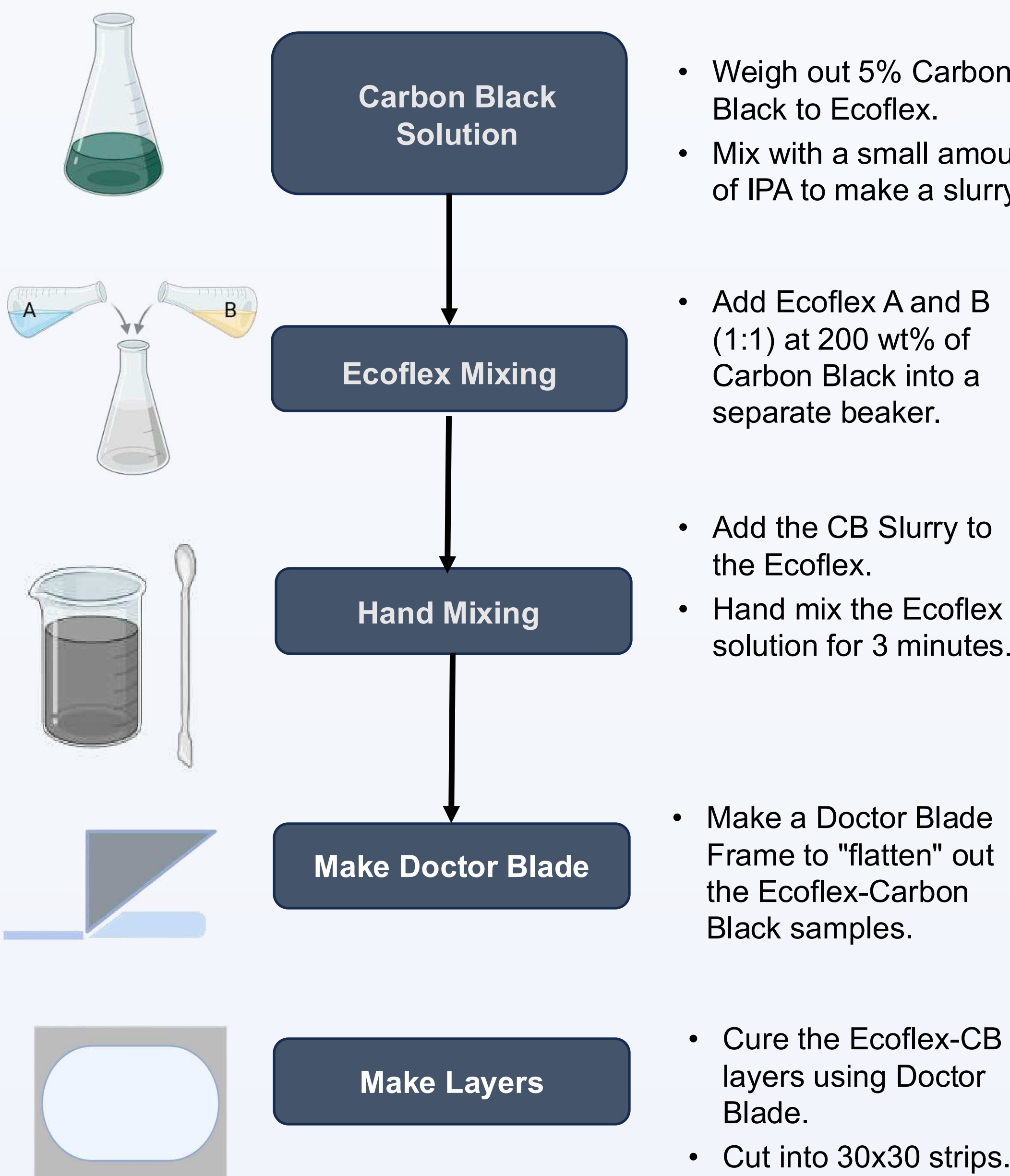
- Independent Variables:** Varying amounts of pre-stretching applied to each electrode layer (Starting from 0% up to 75% in intervals of 26%).
- Dependent Variables:** **Mechanical properties** (Young Modulus, tensile strain, tensile stress), **electrical properties** (resistance), and **DEA performance**.

Justification:

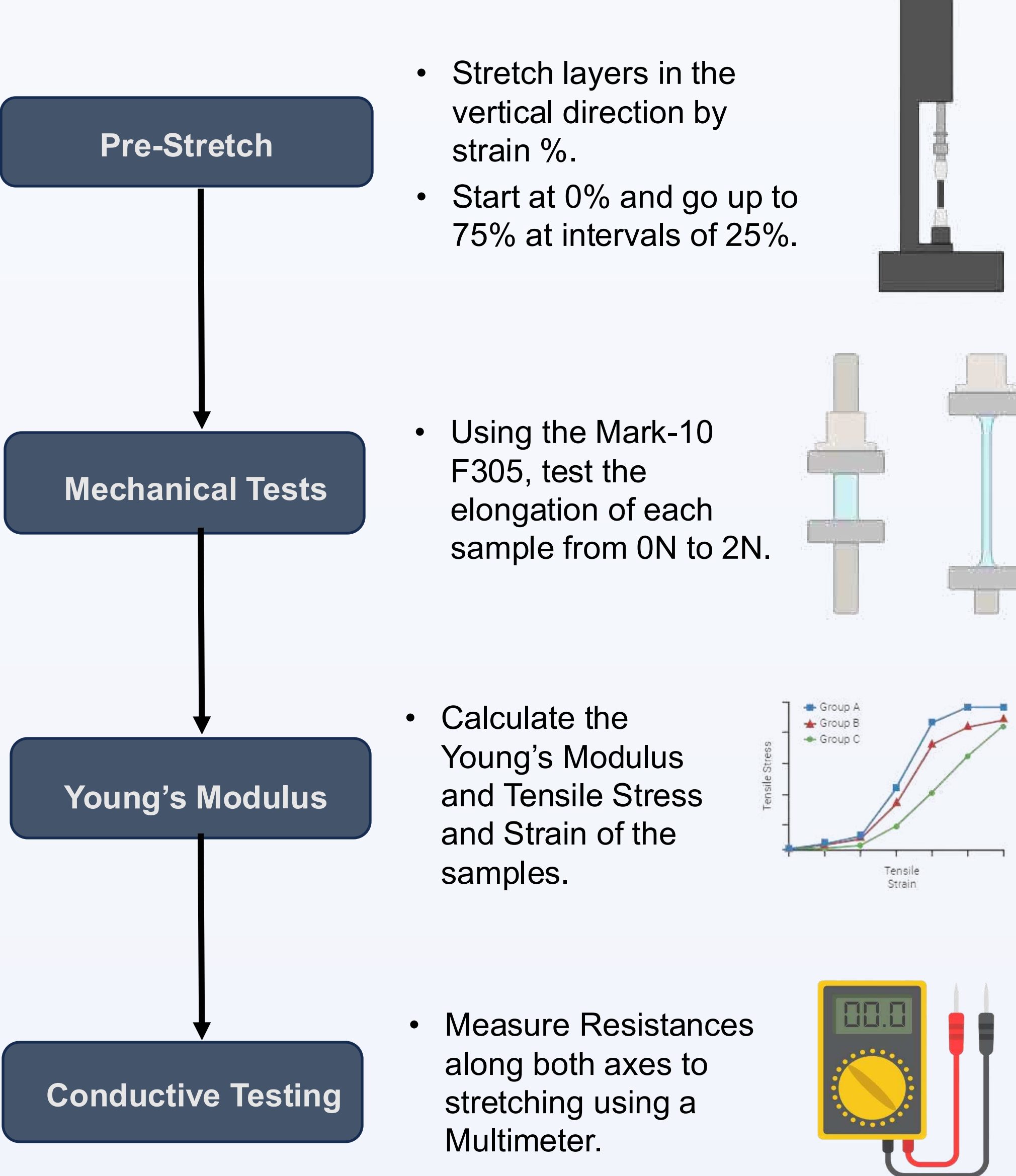
- Previous research on DEAs has only focused on enhancing the dielectric layer, and this project not only focuses on that but also on ADEs, which are more important than DEAs as they have motion in a specific direction.
- Beneficial to research into soft robotics, which are necessary for applications in many fields.

Experimental Design

Making the PDMS Layers



Testing the PDMS Layers



All Images Created in <https://BioRender.com>

Data and Results

Layer #	Stretching (%)	Stretching (mm)	Parallel Resistance (MΩ)				Perpendicular Resistance (MΩ)			
			Trial 1	Trial 2	Trial 3	Average	Trial 1	Trial 2	Trial 3	Average
1	0	0	4.1	3.6	4.8	4.17	4.2	3.8	4.6	4.20
2	25	5	3.9	4.4	3.2	3.83	4.5	3.9	5.2	4.53
3	50	10	3.7	4.2	2.9	3.60	4.8	4.1	5.3	4.73
4	75	15	3.3	4	2.8	3.37	5.1	4.3	5.6	5.00

Table 1: Resistances along Axis Parallel to Stretching and Perpendicular to Stretching.

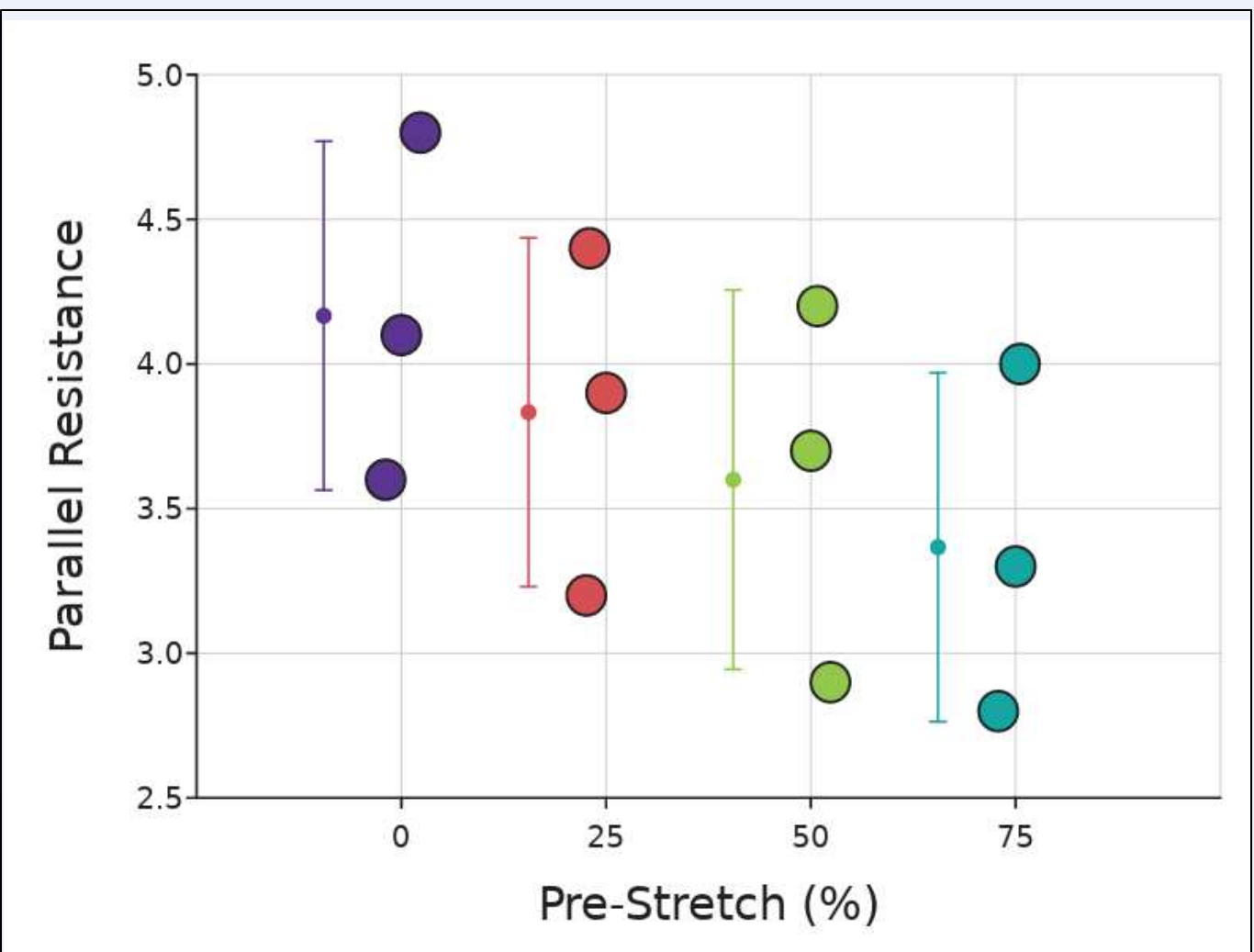


Figure 3. Graph showing Parallel Resistances (MΩ) for each level of stretching (%)

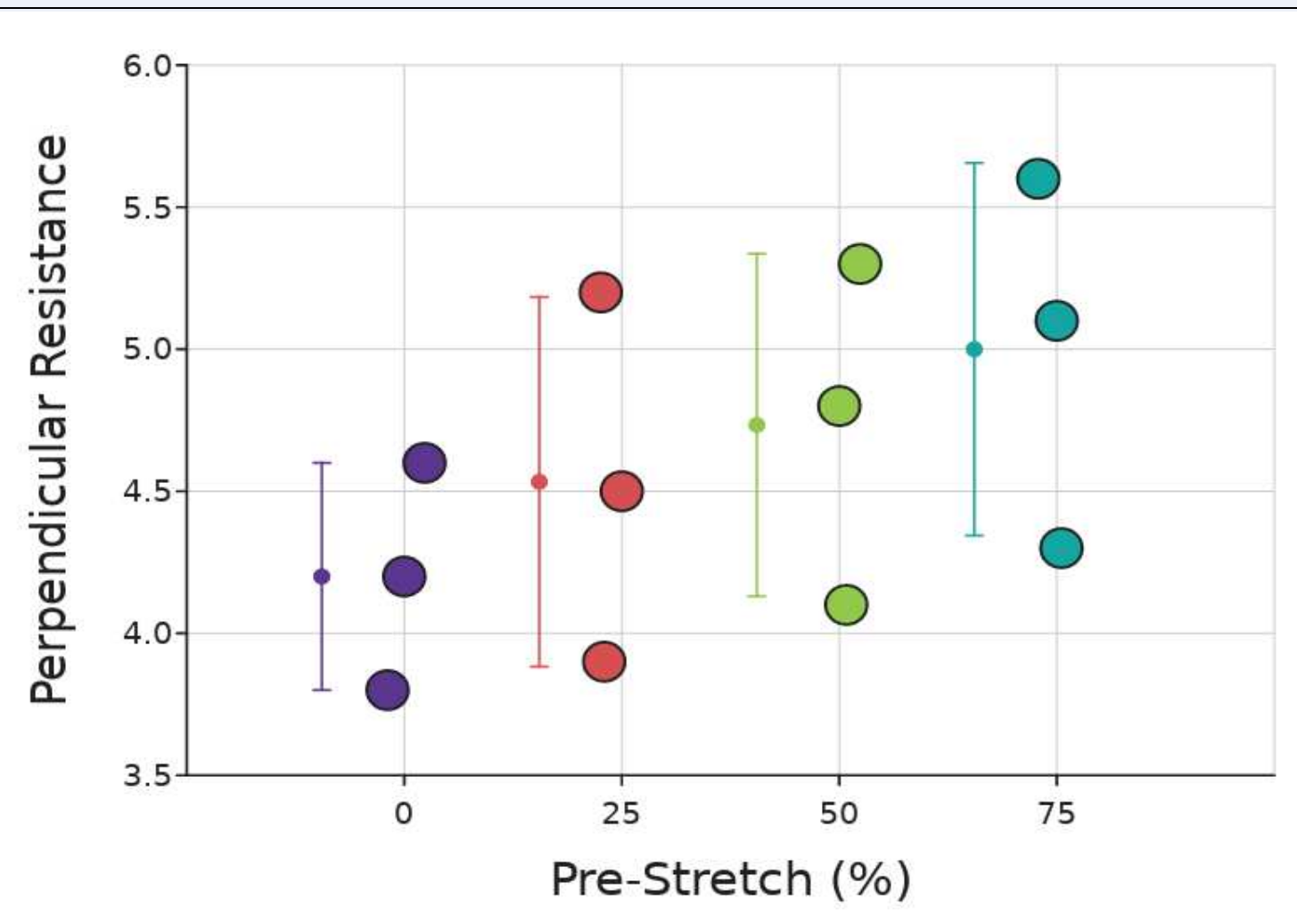


Figure 4. Graph showing Perpendicular Resistances (MΩ) for each level of stretching (%)

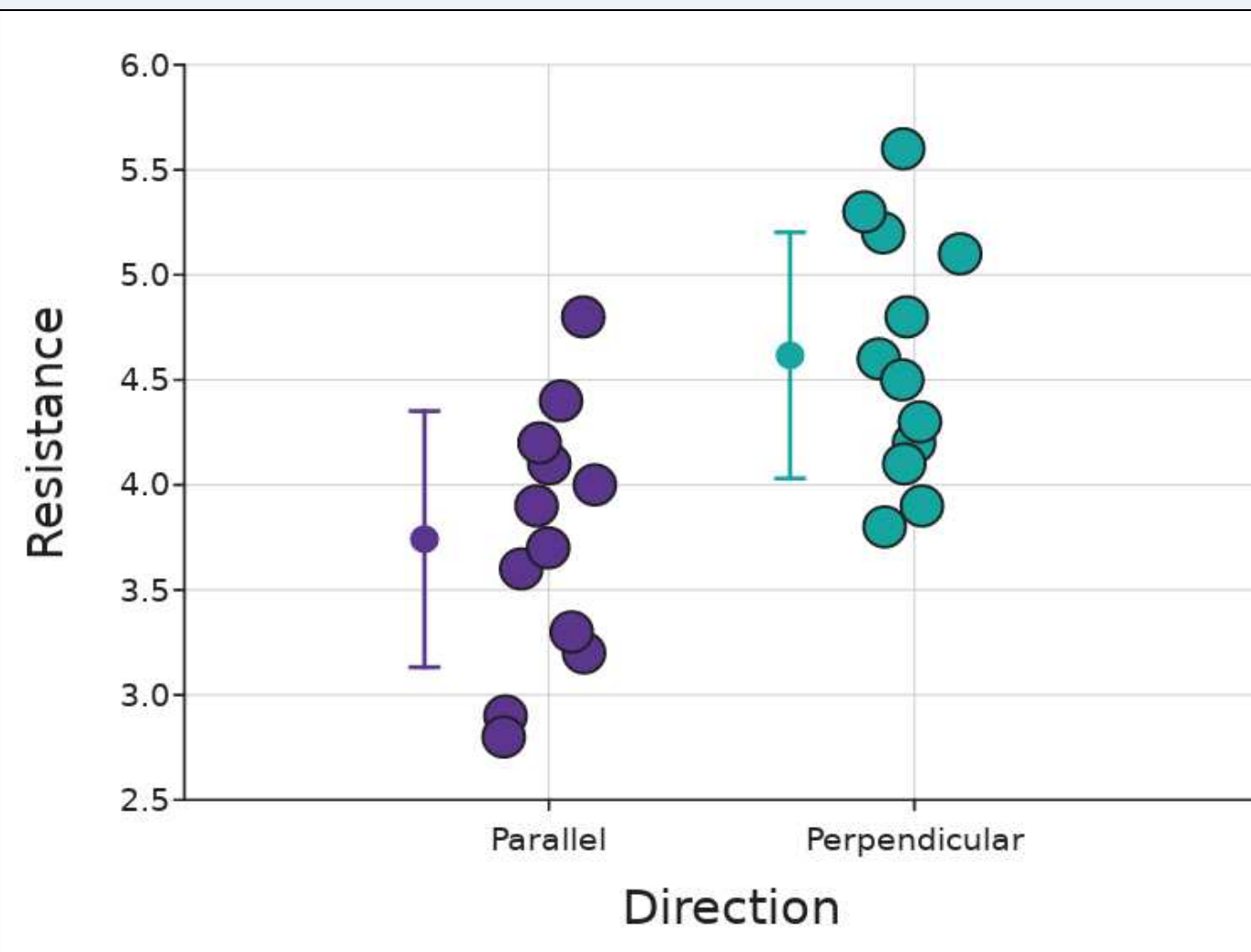


Figure 5. Graph showing all Perpendicular & Parallel Resistances

Conclusions

Observed Trends:

Parallel resistance decreased with pre-stretching (4.17 to 3.37 MΩ) while perpendicular resistance increased (4.20 to 5.00 MΩ), which was consistent with the alignment of Carbon Black particles in the sample along the direction of stretch, which is the goal for ADE designing. The overall divergence along the axes suggest that pre-stretching the electrode layer induces anisotropic electrode behavior, supporting the initial hypothesis.

Statistical Analysis:

However, a Kruskal-Wallis test found that there was no statistically significant difference between the pre-stretch levels for either the parallel or perpendicular resistance, with p-values of 0.52 and 0.43, respectively. Additionally, all comparisons between pairs resulted in p-values ≥ 0.85 . While the observed trends carry physical value, their effect sizes are too small to be confirmed with a sample size of 3 per group.

Broader Implications:

Overall, the results show that pre-stretching shows potential as a potential method for increasing anisotropy in an electrode layer, but more investigation is required. By extending the pre-stretch magnitudes and increasing trial replication, statistical power can be increased.

Future Work

As of now, only conductive data has been collected and processed. For future work, mechanical data needs to be collected and analyzed using the Mark-10 to find the tensile stress/strain and the Young's Modulus. We were unable to do this part of the procedure as the Mark-10 had broken down and will not be fixed for the rest of this year.

References

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